

Iodine Plaque Brachytherapy a Customized Conservative Approach to the Management of Medium/Large-sized Uveal Melanoma

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Abstract. *Background/Aim:* The objective of this study was to assess the role of iodine (¹²⁵I) plaque brachytherapy in the management of uveal melanoma. *Patients and Methods:* This is a retrospective study of 50 patients (median age 67 years; range=33-86 years) with uveal melanoma, treated with ¹²⁵I plaque brachytherapy at the University Hospital of Pisa. Uveal melanoma was diagnosed with A-scan and B-scan standardized echography, fluorescein angiography, indocyanine green-angiography, optical coherence tomography, and/or magnetic resonance imaging. The primary outcomes assessed were local control, overall survival, disease progression, globe preservation, and metastases. Secondary outcomes were acute and late radiation adverse effects. *Results:* Inclusion criteria comprised Eastern Cooperative Oncology Group performance status ≤2, life expectancy >6 months, and tumor thickness ≤10 mm and/or diameter ≤20 mm. All the patients were treated with ¹²⁵I plaque brachytherapy, with a prescription dose of 85 Gy to the tumor apex. The 5-year rate of local control, progression-free survival, metastasis-free survival, enucleation-free survival, and overall survival

were 83.0%, 81.4%, 90.3%, 83.1%, and 92.1% respectively. Twenty-four patients (48.0%) had one or more acute and late toxicities. The most common acute adverse events (CTCAE vs. 5.0) grade 1-3 were conjunctivitis and eye pain (6.0%). Regarding late events, radiation retinopathy grade 1-3 occurred in 18.0% of cases, while grade 1-3 vitreous hemorrhage in 2.5%. *Conclusion:* ¹²⁵I plaque brachytherapy offers an effective and safe approach for selected cases of uveal melanoma, due to the reported satisfactory results in terms of local control, eye conservation and survival.

Uveal melanoma (UM) is the most common primary intraocular malignancy, accounting for 5% of all melanomas (1-3). The median age of diagnosis is approximately 62 years; however, the peak range for diagnosis is between 70 and 75 years (4). UM has a high risk of metastatic progression and an extremely poor prognosis; moreover, local treatment modalities also come with significant visual morbidity (5, 6).

The current accepted standard treatments range from observation to enucleation, depending on the size and the characteristics of the tumor (1-3). In the past years, enucleation represented the standard therapy (7-9) but, over the last four decades, eye sparing treatments with radiotherapy (RT) have assumed a predominant role in the treatment of UM, especially in the form of brachytherapy (BT) or proton beam RT, showing promising results and allowing globe preservation and partial vision sparing.

The collaborative ocular melanoma study (COMS), a randomized trial comparing survival for enucleated patients versus irradiated patients with medium-sized and small choroidal melanomas, demonstrated that BT was equivalent to enucleation in the 5-year overall survival rates and established the role of BT in the treatment of UM (6, 10, 11).

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These data established ^{125}I BT as the gold standard for treating medium-sized uveal melanoma with 5-year risk of local progression of 10.3% (12). Furthermore, several studies suggested that local recurrence was potentially correlated with the risk of metastasis (13).

Currently, eye-preserving, and potentially vision-preserving BT with ^{125}I and ^{106}Ru represents the first-line treatment for UM, since delivery of continuous radiation at a low dose rate minimizes subsequent radiation damage to healthy ocular tissues and therefore reduces the incidence of complications.

With a maximum photon energy of 35 KeV, ^{125}I was used for treating medium and large uveal melanomas, with an apex height of up to 10 mm. The number and location of seeds on each plaque can be adjusted to tailor the dosimetry to each individual tumor.

Using ^{125}I plaques, the generally used radiation dose was 85 Gy, prescribed to the apex or the tumor, as well as the dose used in the COMS study for medium sized UM (11, 14-16). The aim of this retrospective study was to analyze the efficacy and safety of ^{125}I plaques as eye-preserving treatment in UM and to determine prognostic factors for local control (LC), progression-free survival (PFS), metastasis-free survival (MTFS), enucleation-free survival (EFS), and overall survival (OS).

Patients and Methods

A retrospective analysis of patients with UM treated with iodine plaques BT between January 2010 and October 2023 was conducted at the University Hospital of Pisa. The patients were evaluated, treated, and followed by an ocular tumors multidisciplinary team (Ophthalmic Surgeon, Radiation Oncologist and Medical Physicist). Follow up data were collected regarding local control, salvage treatment requirements, occurrence of metastasis, survival status and toxicities.

Inclusion criteria were Eastern Cooperative Oncology Group (ECOG) performance status (PS) ≤ 2 , life expectancy > 6 months, thickness ≤ 10 mm and/or diameter ≤ 20 mm. All the patients had a complete ophthalmologic clinical examination. A-scan and B-scan standardized echography was used to document selected characteristics of tumors at baseline. Criteria were established to assess the consistency of echographic features with the diagnosis of UM. The exhibited features referring to melanoma diagnosis were in conformity with the COMS study: low to medium reflectivity, regular internal structure, a mushroom shape, solid consistency, sound attenuation and internal vascularity (17). Fluorescein angiography and indocyanine green-angiography were used to determine the presence of autonomous circle, lipofuscin, disrupted retinal pigmented epithelium (RPE), and ophthalmoscopy was used to determinate absence of drusen and fibrous metaplasia, to help in the characterization of the UM. Also, optical coherence tomography (OCT) was performed in order to determine irregularity and detachment of the neuroepithelium. A baseline measurement of visual acuity and imaging studies including chest/abdominal/pelvic computed tomography (CT) and/or abdominal ultrasound, magnetic

Table I. *Patients and tumor characteristics.*

Characteristic	Statistics
Age(years)	67 (33-86)
<67	22 (44.0%)
≥ 67	28 (56.0%)
Sex	
Male	28 (56.0%)
Female	22 (44.0 %)
Tumor Side	
Right	23 (46.0 %)
Left	27 (54.0%)
Ciliary body involvement	
No	45 (90.0 %)
Yes	5 (10.0%)
Tumor Size (mm)	
Basal largest diameter	12.0 (5.5-20.0)
Thickness	7.0 (2.6-10.0)
Integrity of the Bruch membrane	
Yes	27 (54.0%)
No	23(46.0%)
AJCC-UICC stage	
1	4 (8.0%)
2	25 (50.0 %)
3	18(36.0 %)
4	3 (6.0 %)

resonance imaging (MRI) and/or whole body FDG PET-CT were required for diagnosis. The eighth edition of the American Joint Committee on Cancer (AJCC) melanoma classification system was used for staging (18). Plaques implants were performed in accordance with American Brachytherapy Society (ABS) recommendations (19).

Based on tumor location and size, different types of plaques were selected for adequate coverage of the tumor and to optimize the placement of the plaque over the tumor. Circular and elliptical episcleral plaques with diameters of 15 and 20 mm were used. They were obtained with positioning of Ioseed ^{125}I sealed sources (Eckert and Ziegler BEBIG GMBH, Berlin, Germany) on customized templates, which were then steam sterilized. A physicist planned a personalized treatment for each patient, based on all available clinical and dimensional findings. An expansion of 2 mm on all sides was applied to the basal dimensions of the tumor. The dose of 85 Gy was prescribed to the tumor apex for lesions greater than 5 mm in height, or to a point located on the central axis 5.0 mm away from the inner sclera for tumors less than 5.0 mm in apical height (12, 16).

Surgery was performed by the Ophthalmic Surgeon in the presence of the radiation oncologist and the medical physicist. Plaque position was determined with a transillumination technique and/or performing binocular indirect ophthalmoscopy, by using a transparent dummy plaque. Muscle transposition was performed whenever necessary.

Plaques were sutured in place and correct positioning was verified for all cases using ultrasound (20). Subsequently, the plaques were left in place for an interval of time depending on the intensity of the source and the prescribed dose and then definitively removed. LC, PFS, MFS, EFS and OS, were examined and calculated using Kaplan-Meier methodology and stepwise regression or the multivariate analysis.

Table II. Local control (LC) in univariate and multivariate analysis.

LC Factor	Univariate analysis			Multivariate analysis		
	HR	95%CI	p-Value	HR	95%CI	p-Value
Age (0) <67; (1) $\geq$ 67	2.339	0.707-7.738	0.1639			
Sex (0) M; (1) F	0.570	0.127-2.546	0.570			
Tumor side (0) Left; (1) Right	0.354	0.071-1.763	0.205			
Ciliary body involvement (0) No (1) Yes	3.454	0.965-13.017	0.0567			
Largest Basal diameter (mm) (0) <12; (1) $\geq$ 12	1.115	0.921-1.348	0.2644			
Thickness (mm) (0) <7; (1) $\geq$ 7	1.250	1.060-1.474	0.0079	1.258	1.048-1.510	0.0136
Integrity of the Bruch membrane (0) No (1) Yes	0.045	0.707-13.475	0.642			
Tumor stage (0) 1+2; (1) 3+4	3.162	0.607-16.475	0.172			

LC: Local control; PFS: progression-free survival; HR: hazard ratio; 95%CI: 95% confidence interval. Significant differences are shown in bold.

Clinical evaluation was performed by Radiation Oncologists and Ophthalmologists one week after radiotherapy and every three months from the end of surgery in the first year, and every six months for five years and every twelve months subsequently. Acute toxicities were defined as symptoms occurring within three months after treatment completion, whereas late toxicities were defined as symptoms developing after three months. Radiation induced toxicities were scored according to Common Terminology Criteria for Adverse Events [CTCAE v 5.0] (21). All the patients were observed until they died or until October 2023. The median follow-up of survivors was 73.6 months (range=3-165 months).

Statistical analysis. LC, PFS, MTFS, EFS and OS were assessed as end-points, using the following covariates: patient age, sex, eye laterality, largest tumor diameter, height of the tumor, site, integrity of the Bruch membrane, tumor stage. Survival curves were calculated using the Kaplan-Meier method and log-rank test was applied to evaluate differences between curves. Covariates that influenced survival ($p < 0.1$) after univariate analysis were included in a Cox regression model for multivariate analysis. The results of survival analysis are expressed using hazard ratios (HR) with 95% confidence interval (95%CI). Significance was set at 0.05 and all analyses were carried out using SPSS v.27 technology (IBM, Armonk, NY, USA).

Results

Patient characteristics are shown in Table I. All the patients were treated with I^{125} plaques with a prescription dose of 85 Gy to the tumor apex. An accurate ophthalmic examination three months after BT showed LC of disease in 45 (90.0%) patients. Patients with lesions with a height less than or equal to 7 mm had a significantly higher LC probability when

compared to those with greater height (HR=1.25, 95%CI=1.06-1.47, $p=0.007$); these findings were confirmed in multivariate analysis (HR=1.25, 95%CI=1.04-1.51, $p=0.013$). Moreover, patients who showed ciliary body involvement had a trend of higher LC, (HR=3.45, 95%CI=0.96-13.01, $p=0.056$); these data were not confirmed in multivariate analysis (Table II).

Patients with lesions with ciliary body involvement had a significantly higher risk of progression of disease when compared with those without ciliary body involvement, (HR=3.52, 95%CI=0.95-12.94, $p=0.005$); however, the data were not confirmed in multivariate analysis. Patients with lesions with a height less than or equal to 7 mm had a significantly higher PFS probability when compared with those with higher height (HR=1.25, 95%CI=1.06-1.47, $p=0.007$); data showed a trend in multivariate analysis (HR=8.43, 95%CI=2.17-30.20, $p=0.057$). Moreover, enucleated patients had higher risk of progression in univariate and multivariate analysis ($p=0.001$ and $p=0.002$, respectively) (Table III).

Distant metastases were recorded in four patients (8.0%); two of them (4.0%) had also local relapse. All the patients with distant disease showed liver metastasis and underwent immunotherapy. Patients with lesions with ciliary body involvement had a significantly higher risk of progression of distant disease when compared with those without ciliary body (HR=32.01, 95%CI=3.30-31.54, $p=0.002$); data were confirmed in multivariate analysis (HR=25.65, 95%CI=1.95-36.93, $p=0.013$). Patients with lesions with basal diameter less than or equal to 12 mm had a trend for a lower distant

Table III. Progression-free survival (PFS) in univariate and multivariate analysis.

PFS Factor	Univariate analysis			Multivariate analysis		
	HR	95%CI	p-Value	HR	95%CI	p-Value
Age (0) <67; (1) ≥67	2.348	0.710-7.766	0.1620			
Sex (0) M; (1) F	0.699	0.345-1.414	0.319			
Tumor side (0) Left; (1) Right	0.745	0.091-1.866	0.305			
Ciliary body involvement (0) No (1) Yes	3.522	0.959-12.941	0.0579			
Largest Basal diameter (mm) (0) <12; (1) ≥12	1.111	0.920-1.341	0.2704			
Thickness (mm) (0) <7; (1) ≥7	1.250	1.060-1.473	0.0079	2.528	0.978-9.9320	0.0572
Integrity of the Bruch membrane (0) No (1) Yes	0.075	0.907-23.475	0.746			
Tumor stage (0) 1+2; (1) 3+4	3.162	0.607-16.475	0.172			
Enucleation (0) No (1) Yes	12.843	4.053-40.696	0.0010	8.344	2.155-32.302	0.0021

HR: Hazard ratio; 95%CI: 95% confidence interval. Significant differences are shown in bold.

Table IV. Metastasis-free survival (MFS) in univariate and multivariate analysis.

MFS Factor	Univariate analysis			Multivariate analysis		
	HR	95%CI	p-Value	HR	95%CI	p-Value
Age (0) <67; (1) ≥67	0.7287	0.000-2.541	0.9569			
Sex (0) M; (1) F	0.699	0.345-1.414	0.319			
Tumor Side (0) Left; (1) Right	0.745	0.091-1.866	0.305			
Ciliary body involvement (0) No (1) Yes	32.019	3.301-31.546	0.0028	25.658	1.954-36.937	0.0135
Largest Basal diameter (mm) (0) <12; (1) ≥12	1.381	0.9979-1.911	0.0515			
Thickness (mm) (0) <7; (1) ≥7	1.050	0.767-1.437	0.7608			
Integrity of the Bruch membrane (0) No (1) Yes	0.078	0.607-11.475	0.9459			
Tumor stage (0) 1+2; (1) 3+4	1.828	0.494-1.343	0.3664			
Local control (0) No (1) Yes	0.566	0.000-6.769	0.3664			

HR: Hazard ratio; 95%CI: 95% confidence interval. Significant differences are shown in bold.

spread when compared with those with longer basal diameter (HR=1.38, 95%CI=0.99-1.91, $p=0.051$).

Moreover, patients who underwent enucleation had a higher risk of distant progression in univariate analysis ($p=0.068$) (Table IV). Enucleation was required in six patients (12.0%):

three (6.0%) for failure of LC, three (6.0%) for late toxicities (all three eyes developed a severe unresponsive refractory neovascular glaucoma). Patients with lesions with ciliary body involvement tended to have a higher risk of disease progression when compared with those without ciliary body involvement

Table V. *Enucleation-free survival (EFS) in univariate and multivariate analysis.*

EFS Factor	Univariate analysis			Multivariate analysis		
	HR	95%CI	p-Value	HR	95%CI	p-Value
Age (0) <67; (1) ≥67	0.9605	0.188-4.900	0.9614			
Sex (0) M; (1) F	0.399	0.445-1.614	0.519			
Tumor side (0) Left; (1) Right	0.845	1.081-2.966	0.507			
Ciliary body involvement (0) No (1) Yes	5.188	0.947-28.424	0.0578	12.192	1.092-136.170	0.042
Largest Basal diameter (mm) (0) <12; (1) ≥12	1.133	0.871-1.474	0.3516			
Thickness (mm) (0) <7; (1) ≥7	1.377	1.094-1.733	0.0064	1.511	1.031-2.214	0.034
Integrity of the Bruch membrane (0) No (1) Yes	0.078	0.607-11.475	0.9458			
Tumor stage (0) 1+2; (1) 3+4	3.718	1.146-12.062	0.0288			

HR: Hazard ratio; 95%CI: 95% confidence interval. Significant differences are shown in bold.

Table VI. *Overall survival (OS) in univariate and multivariate analysis.*

OS Factor	Univariate analysis			Multivariate analysis		
	HR	95%CI	p-Value	HR	95%CI	p-Value
Age (0) <67; (1) ≥67	0.829	0.000-2.164	0.9541			
Sex (0) M; (1) F	0.734	0.318-1.694	0.468			
Tumor side (0) Left; (1) Right	0.587	0.198-1.741	0.337			
Ciliary body involvement (0) No (1) Yes	6.954	1.680-28.784	0.0074			
Largest Basal diameter (mm) (0) <12; (1) ≥12	1.117	0.891-1.401	0.3382			
Thickness (mm) (0) <7; (1) ≥7	1.098	0.913-1.320	0.3202			
Integrity of the Bruch membrane (0) No (1) Yes	1.541	0.359-6.613	0.560			
Tumor stage (0) 1+2; (1) 3+4	1.560	0.612-3.982	0.3517			
Enucleation (0) No (1) Yes		0.644-10.370	0.1807			

HR: Hazard ratio; 95%CI: 95% confidence interval. Significant differences are shown in bold.

(HR=5.18, 95%CI=0.94-28.42, $p=0.057$); data were confirmed in multivariate analysis, with a statistical significance (HR=12.19, 95%CI=1.09-13.17, $p=0.042$). Patients with lesions with a height more than 7 mm had a trend for higher probability of enucleation when compared with those with higher height (HR=1.37, 95%CI=1.09-1.73, $p=0.064$); data

were confirmed in multivariate analysis, with statistical significance (HR=1.51, 95%CI=1.03-2.21, $p=0.034$). Patients with lesions of higher stage (3-4) had significantly higher probability of enucleation when compared with those with lower stage (1-2) (HR=3.71, 95%CI=1.14-12.06, $p=0.028$); data were not confirmed in multivariate analysis (Table V).

Table VII. Acute and late toxicities after iodine ¹²⁵ brachytherapy for uveal melanoma.

	G1	G2	G3	G4	
Acute toxicity					
Eyelid function disorder	1 (2.0%)	0	0	0	1 (2.0%)
Eye pain	3 (6.0%)	0	0	0	3 (6.0%)
Watering eyes	1 (2.0%)	0	0	0	1 (2.0%)
Photophobia	0	0	0	0	0
Conjunctivitis	2 (4.0%)	1 (2.0%)	0	0	3 (6.0%)
Extraocular muscle disorder	0	0	1	0	1 (2.0%)
Late toxicity					
Cataract	0	0	3 (6.0%)	0	3 (6.0%)
Loss visual acuity	8 (16.0%)	10 (20.0%)	0	0	18 (36.0%)
Diplopy	5 (10.0%)	0	0	0	5 (10.0%)
Neovascular glaucoma	0	2	0	0	2 (4.0%)
Exudative retinal detachment	0	2	0	0	2 (4.0%)
Radiation retinopathy	1 (2.0%)	6 (12.0%)	2 (4.0%)	0	9 (18.0%)
Vitreous hemorrhage	2 (4.0%)	1 (2.0%)	5 (10.0%)	0	8 (16.0%)

G1: Grade 1; G2: grade 2; G3: grade 3; G4: grade 4.

Nine patients (18.0%) died during the observation time: four (8.0%) because of distant progression, and the other five (10.0%) because of old age and/or comorbidities. Patients with lesions in the ciliary body had significantly higher risk of death when compared with those without ciliary body involvement (HR=6.95, 95%CI=1.68-28.78, $p=0.007$); data were not confirmed in multivariate analysis (Table VI).

Patients subject to enucleation had no difference in survival compared to patients not treated with demolitive surgery; LC was 83.0% and 59.6%, PFS was 81.4% and 58.5%, MFS was 90.3% and 66.8%, EFS was 83.1% and 59.5% and OS was 92.1% and 59.5% at 5 and 10 years, respectively. Overall acute and late toxicities are reported in Table VII.

Visual acuity was <20/200 in eight patients (16.0%); a progressive reduction of at least three Snellen lines was recorded in 18 patients (36.0%). Twenty-four patients (48.0%) had one or more acute and late toxicities. The most frequent common acute adverse events (CTCAE vs. 5.0) grade 1-2 were conjunctivitis and eye pain (6.0%, both) the only grade 3 acute toxicity was extraocular muscle disorder (2.0%). Regarding late events the most recurrent were grade 1-3 radiation retinopathy (18.0%), and grade 1-3 vitreous hemorrhage (2.5%).

Discussion

This retrospective study evaluated a cohort of patients with UM treated with iodine plaques BT and outlined the predictor factors of local control, salvage treatment

requirements, occurrence of metastasis, survival status and toxicities. In the current study, the 5-year overall survival of 92.1% was higher than that observed in the COMS study (81.0%) (11). Moreover, in the COMS study, the five-year rate of death with UM metastasis was 9.0% in the BT arm, while in our study only four (8%) of the nine patients died from liver metastasis: two (4.0%) died for liver metastasis, two (4.0%) of local disease and liver metastasis and three (6.0%) of local progression and two (4.0%) of other causes.

One of the most important clinical features for estimation of UM prognosis is tumor size (22-24). Tumor size is most often measured as chord or arc length of largest basal diameter and tumor thickness at the apex of the tumor. In our study the highest thickness at the apex of the tumor was an important clinical and pathological prognostic factor, being a statistically significant factor for the measured parameters: LC, PFS and EFS ($p=0.079$, $p=0.079$ and $p=0.064$ at the univariate analysis, respectively; and $p=0.013$, $p=0.057$ and $p=0.034$ at the multivariate analysis, respectively). We showed that the prognosis is relatively better when the thickness is 7.0 mm or less. Shields *et al.* (25) evaluated prognosis based on precise melanoma thickness in over 8,000 cases and found that each increase by one millimeter increased also the risk for metastasis by 5%. In our study having a tumor with a thickness ≥ 7.0 mm means having a risk of 1.2 greater of that of not having LC, this risk was confirmed by the multivariate analysis. Moreover, for each mm increase in tumor thickness, the risk was 1.3 times higher of undergoing enucleation.

With regard the largest basal diameter, we found a trend of association with the increasing risk for MFS ($p=0.051$); in the same way in the COMS study the only factors correlated with death from metastasis and all-cause mortality were age and maximum basal diameter (11).

In the study of Damato *et al.*, 76 patients died (67 from metastasis). Multivariate analysis showed that the most significant factors of death were basal tumor diameter ($p<0.001$), monosomy 3 ($p<0.001$), and epithelioid cellularity ($p=0.004$) (26). Further theoretical analysis estimated that micrometastasis occur at an approximate tumor volume of 7 mm³, which corresponds to a tumor of 3 mm diameter and 1.5 mm thickness when is clinically visible (27).

We showed that the tumor location had indeed a prognostic value. As previously mentioned, UM can arise in the iris, ciliary body, or choroid. Iris melanoma has a better prognosis and ciliary body melanoma has the worst prognosis; the management of ciliary body melanoma is controversial (25, 28-30). In a cohort of 8,033 patients, Shields *et al.* found that the metastasis rates at 5 and 10 years were 4% and 7% for iris melanoma, 19% and 33% for ciliary body melanoma, and 15% and 25% for choroidal melanoma, respectively (25). In this study we showed that the involvement of the ciliary body was statistically significant for all the prognostic variables, in univariate and univariate analyses. Several authors explained that tumors involving the ciliary body have a predilection for two chromosomal abnormalities (monosomy of 3 and multiple copies of 8q) that were associated with a poor prognosis (31, 32). Tumors with ciliary body involvement also express vascular patterns that are correlated with decreased patient survival (29, 32, 33). According to our study, patients with UM with involvement of ciliary body had a lower survival ($p=0.74$). Also, these patients had 5.18 times higher risk of undergoing enucleation and 32.01 times higher risk of metastasis than those without ciliary body involvement. Li *et al.* reported tumors with 100% ciliary body involvement had 3.6 times higher risk of metastasis than choroidal melanoma (29). However, in our study, ciliary body involvement was only associated with a trend for the LC and PFS ($p=0.056$, $p=0.057$, respectively). We concluded that ciliary body involvement was an independent predictor of survival in several multivariate models (28, 29-34).

In our analysis, patients with AJCC Stage 3-4 had 3.71 times higher risk of undergoing enucleation. Moreover, patients who underwent enucleation had 12.84 times higher risk for lower PFS in univariate analysis and 8.34 times higher risk of lower PFS in multivariate analysis ($p=0.001$ and $p=0.002$, respectively).

We found that 24 patients (48.0%) had one or more acute and late toxicities. As seen in Table VII, the most frequent common acute adverse event (CTCAE vs. 5.0) grade 1-2 were conjunctivitis and eye pain (6.0%, respectively); the

only grade 3 acute toxicity was extraocular muscle disorder (2.0%). Regarding late events the most recurrent were grade 1-3 radiation retinopathy (18.0%), and grade 1-3 vitreous hemorrhage (2.5%). The goals of radiotherapy treatment for cancer are to achieve tumor control/destruction, while minimizing damage to healthy tissue (35, 36).

Due to different methods of dosimetry and tumor characteristics in several studies, it is difficult to compare the rate of complications between different cohorts (37, 38). In a cohort of patients with tumor apex height ≤ 5 mm, visual complications correlated with retinopathy ($p=0.03$) and cataract ($p<0.01$) from ruthenium plaques BT were less severe than those recorded from iodine plaques BT (39-42).

We found a neovascular glaucoma rate of 4.0%, lower than that reported in a retrospective review over 10-years (6.7-8.6%) (40). Thanks to the growing role of the multidisciplinary management of UM, there was a shift toward globe-sparing techniques, with radioactive plaques BT, charged particles, and photons. BT with $I-125$ and $Ru-106$ has achieved a vast acceptance and has become the most common form of radiotherapy for patients affected by primary UM, with a good LC and a reduced incidence of complications (43). However, BT limitations were tumor size (>10 mm thickness, >16 mm diameter) or tumor location (lesions near the optic nerve and in the posterior lobe of the eye); thus, in these cases, other conservative treatments such as proton-beam or stereotactic photon-beam RT have been suggested. Stereotactic body radiotherapy (SBRT) is administered using a few large fractions or a single large fraction (SRS). In our experience one large single fraction (27 Gy) of stereotactic RT can be considered a useful therapeutic approach for UM, also because melanoma is considered a radioresistant tumor (44).

Conclusion

Local treatment with ^{125}I BT for primary UM is effective in preventing local recurrence in more than 83.0% of cases at five years after treatment, with a low risk of metastasis in patients with localized UM and depending on tumor stage/size at diagnosis. ^{125}I BT is also a conservative and functional eye sparing therapeutic approach. As an alternative to enucleation, ^{125}I BT, used depending on lesion thickness, preserves the eye and vision, and offers excellent local control rates and cosmetic outcomes. We showed that prognostic factors are essential for predicting patient outcome; therefore, a multidisciplinary approach is strongly suggested.

Conflicts of Interest

All the Authors declare no conflicts of interest in relation to this study.

Authors' Contributions

Conception: C. Laliscia and F. Genovesi- Ebert. Study design: C. Laliscia, F. Genovesi- Ebert and F. Paiar. Data collection and processing: T. Fuentes, F. Cresti, F. Guido and C. Trippa. Article writing: C. Laliscia. Critical review: F. Genovesi- Ebert, F. Perrone, F. Paiar and G. Pellegrini. Performed data collection, statistical analysis: F. Perrone. Funding Acquisition, G. Pellegrini, and F. Genovesi-Ebert.

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